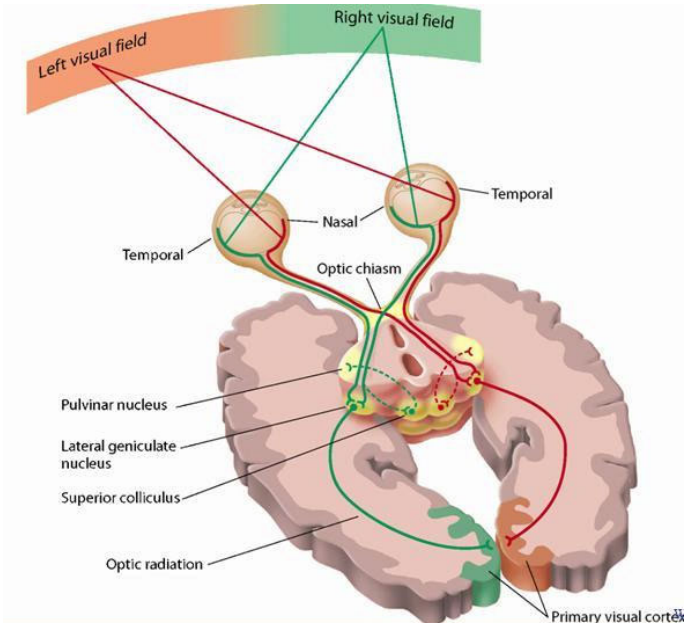
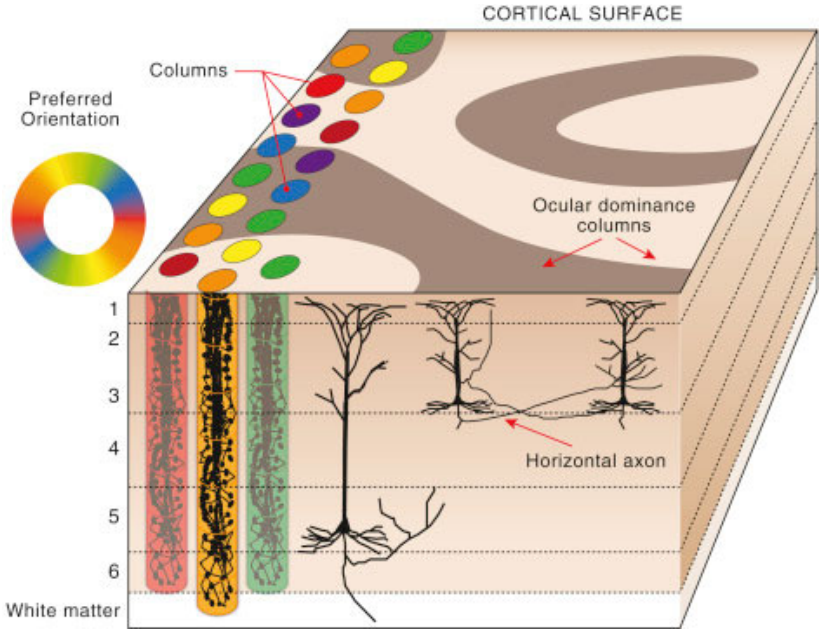
NEUR 416 Final Project

PRESENTED ON APRIL 25, 2025 BY JON FLYNN

NEUR 416





Let’s build an orientation column!

And then measure how good it is

What is the first component needed to establishes cortical shape?

Building the Stimuli Set

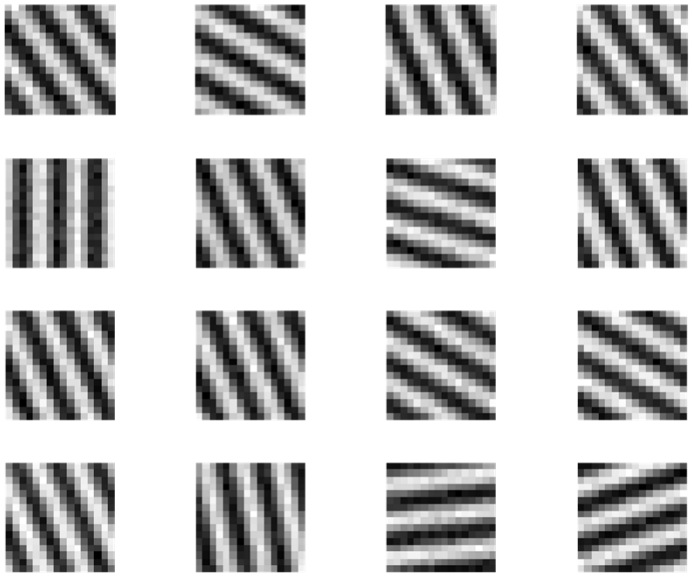
Give 16 evenly spaced *different* orientations from 0 to 180

Add a little bit of noise (between .01 and .1)

Random phase, 3 cycles per image

40,000 training trials

- Please do not make a 16x16x40,000 matrix



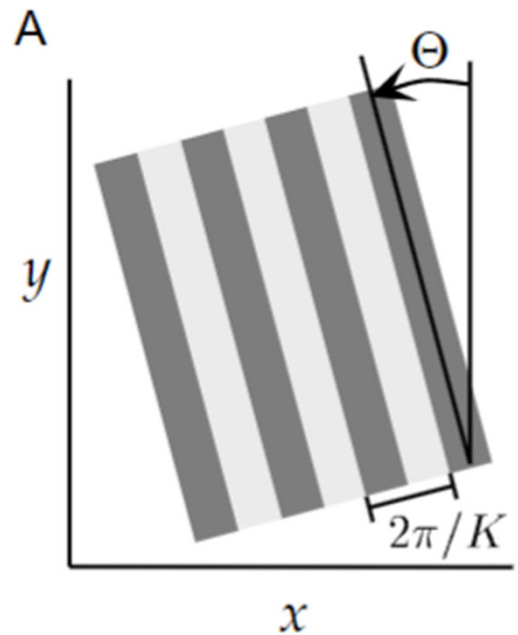
Generating Contrast Gratings

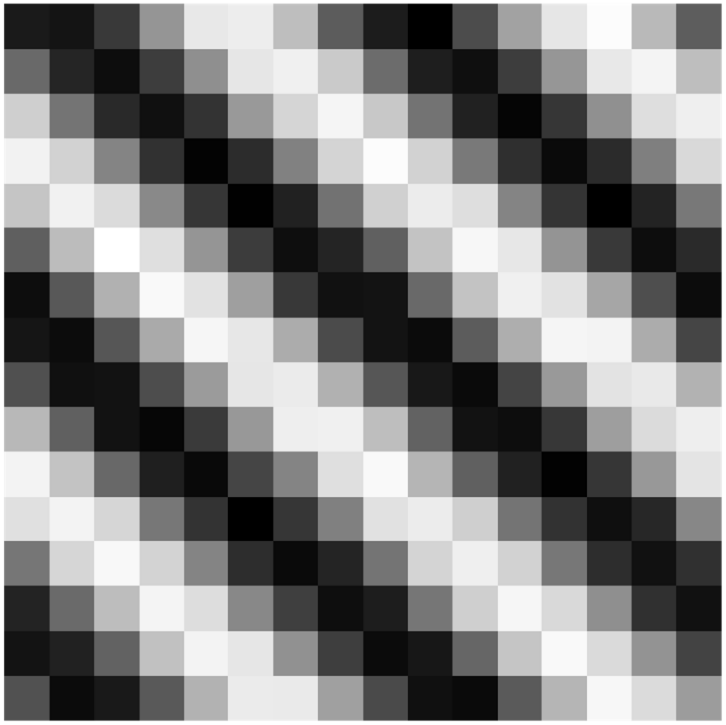
$s(x, y) = A\cos(Kx\cos(\Theta) + \mathrm{Ky}\sin(\Theta) - \Phi)$

Where
- A is the amplitude
- K is the spatial frequency
- Θ is the orientation angle
- Φ is the phase (randomly selected)

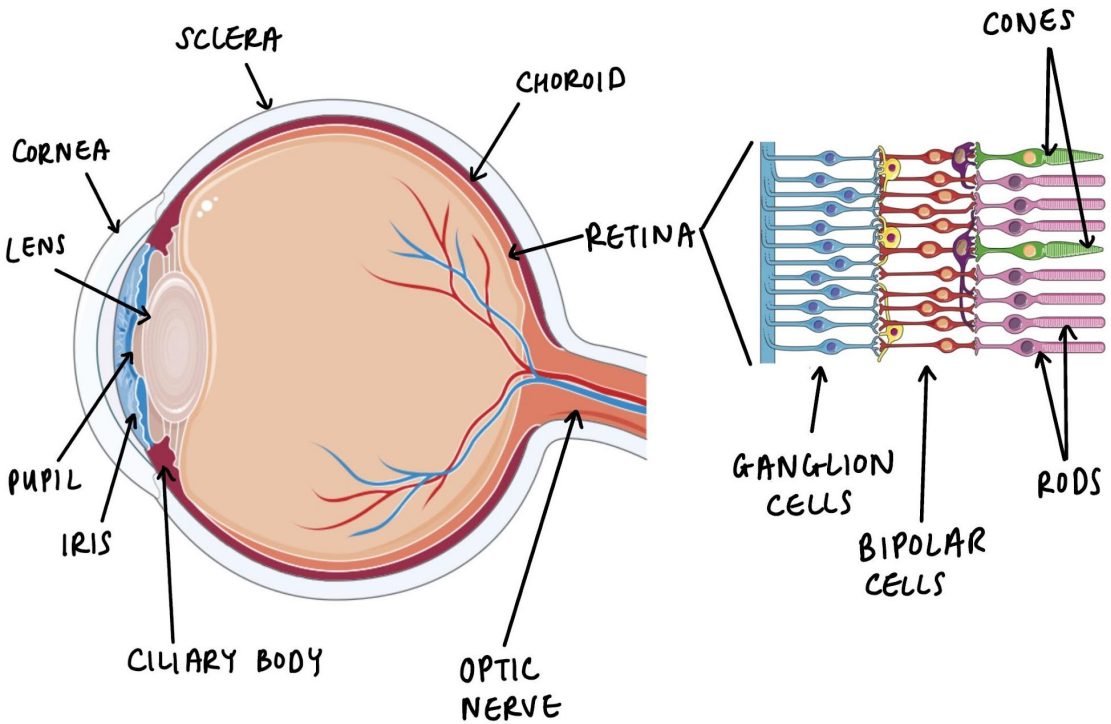
Hint: Use meshgrid to set up initial screen grating

How would we add random noise to this?
- noiseSD*randn(imgSz);





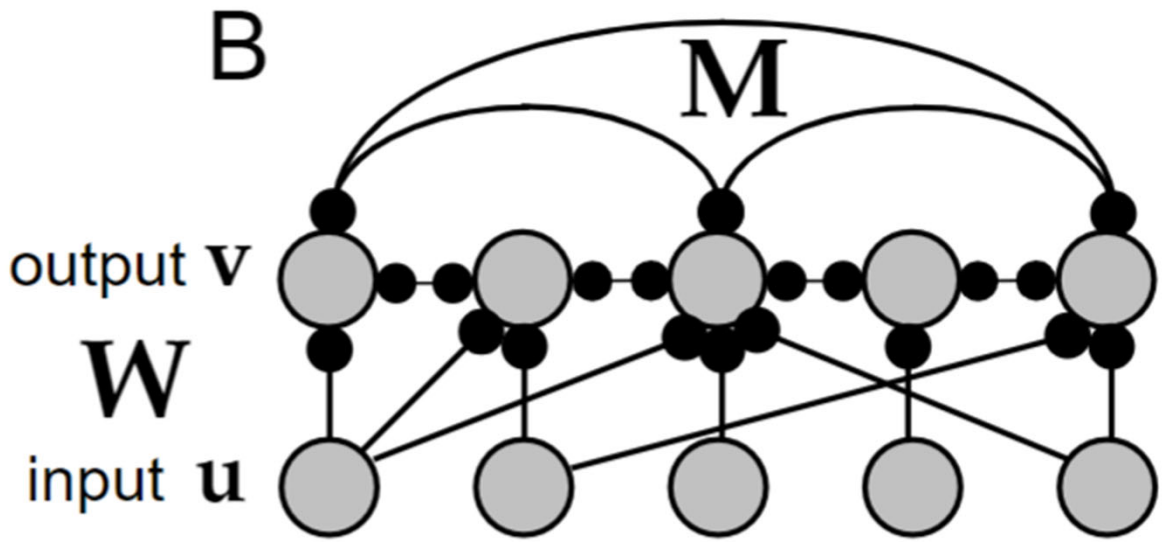
How do we turn a 16x16 orientation grating into an input?



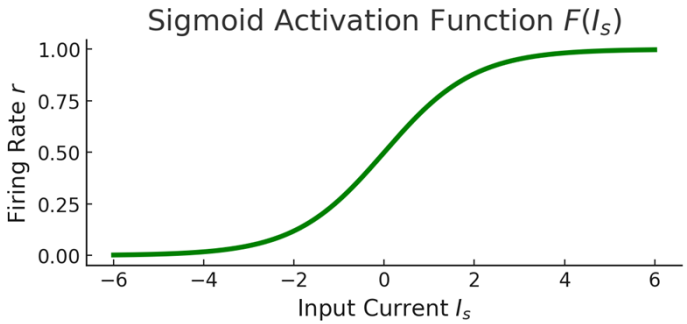
Normalizing the input is an option, but shouldn't be necessary here

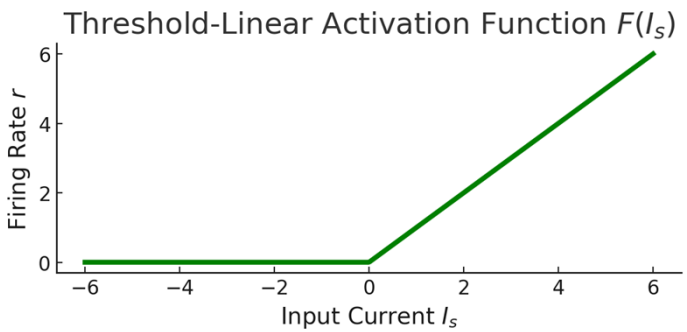
```
u = u - mean(u);
u = u / (std(u)+eps);
```

Good for removing useless constant brightness



Computing the Output





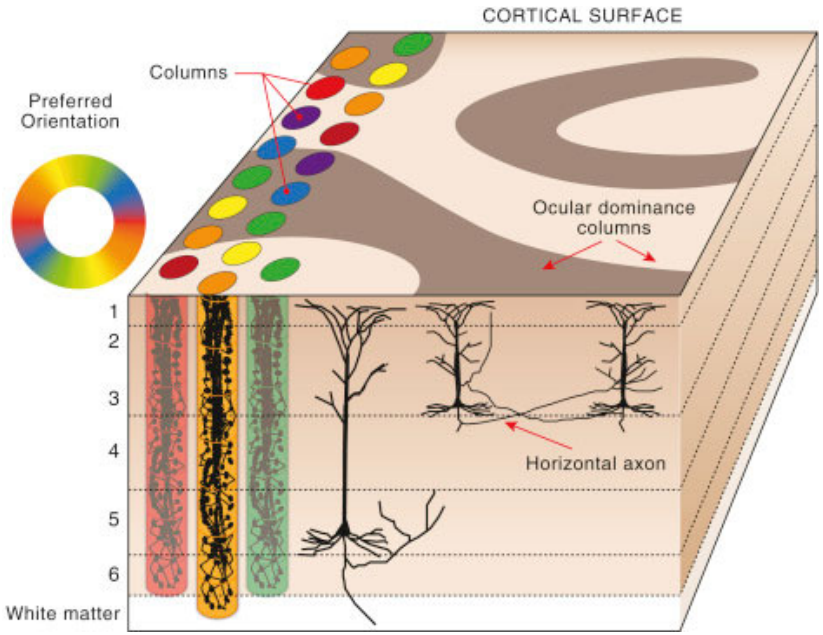
<u>Recall:</u> What might make a good activation function (F)?

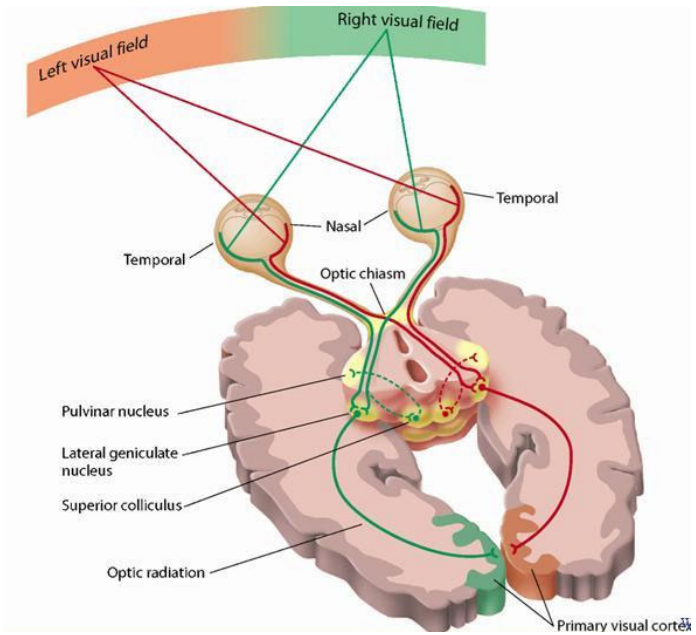
$$v = F(I_s)$$

Effectively converts nA to Hz

Sigmoids are bounded and easily analyzed

To capture thresholding behavior very simply, rectified linear functions like $[I_s - \gamma]_+$ are used

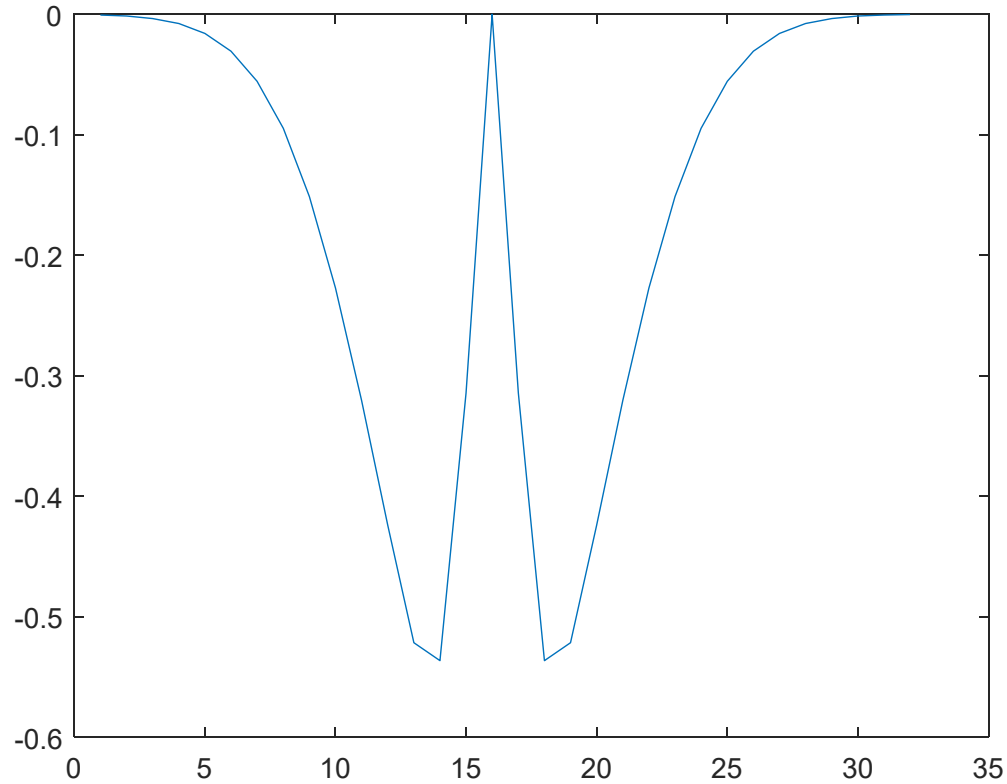




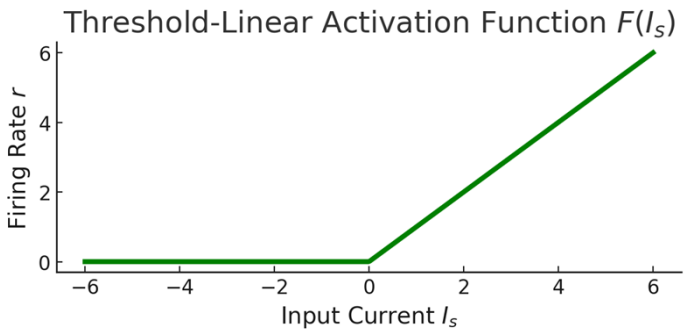
Orientation columns display mid range inhibition (and lots else)

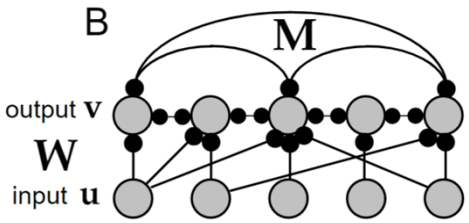
Lateral Inhibition Function

Also called the Ricker Wave or Mexican Hat function



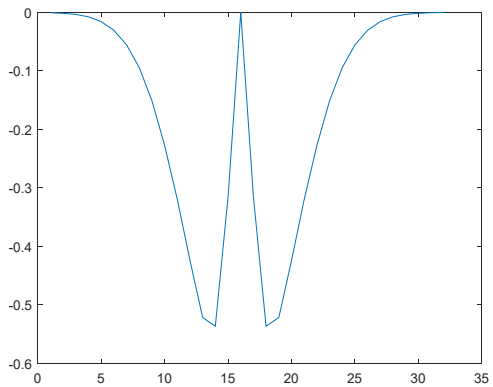
$$v_a^{\text{new}} = \text{ReLU}\left(v_a^{\text{old}} + \sum_{b \neq a} M_{ab} \cdot v_b^{\text{old}}\right)$$





Network Structure

Simple activation function (sigmoid or Linear Activation)



Single lateral inhibition step each trial

Remember to apply a linear threshold after the inhibition!

Training

40,000 TRIALS OF STIMULI RANDOMLY SELECTED FROM OUR SET OF 16 ORIENTATION

Modifying Hebbian Plasticity to a subtractive normalization rule

$$\tau_w \frac{d\boldsymbol{w}}{dt} = v\boldsymbol{u} - \frac{v(\boldsymbol{n} \cdot \boldsymbol{u})\boldsymbol{n}}{N_u}$$

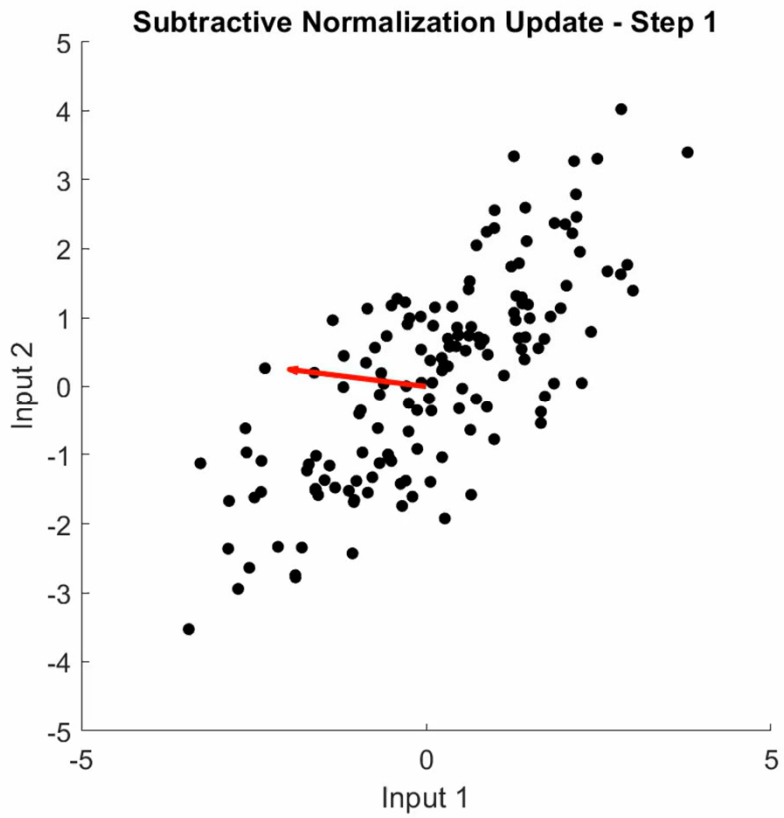
Where

- N_u is the number of all inputs
- $\boldsymbol{n}$ is an N_u dimension vector of all ones
- $\boldsymbol{n} \cdot \boldsymbol{u}$ is the summation of all inputs

Magic happens if $\boldsymbol{e_1} \propto \boldsymbol{n}$

$$\boldsymbol{w}(t) = (\boldsymbol{w}(0) \cdot \boldsymbol{e_1})\boldsymbol{e_1} + \sum_{\mu=2}^{N_u} \exp\left(\frac{\lambda_\mu t}{\tau_w}\right)(\boldsymbol{w}(0) \cdot \boldsymbol{e_\mu})\boldsymbol{e_\mu}$$

Subtractive Normalization for the second principal component



What was the problem with subtractive normalization?

Implementing a bounding procedure

For all the weights into a given output, take the mean and subtract from the weight to enforce a zero-mean weight for each output neuron a

$$W_{ab} \leftarrow W_{ab} - \frac{1}{\mathrm{N}_{\mathrm{input}}} \sum_{b} W_{ab}$$

For all the weights into a given output, divide out the L2 norm

$$W_a \leftarrow \frac{W_a}{\sqrt{\sum_b W_{ab}^2}}$$

Include *eps* in calculation, as $W \rightarrow 0$ often

Quick verification after training

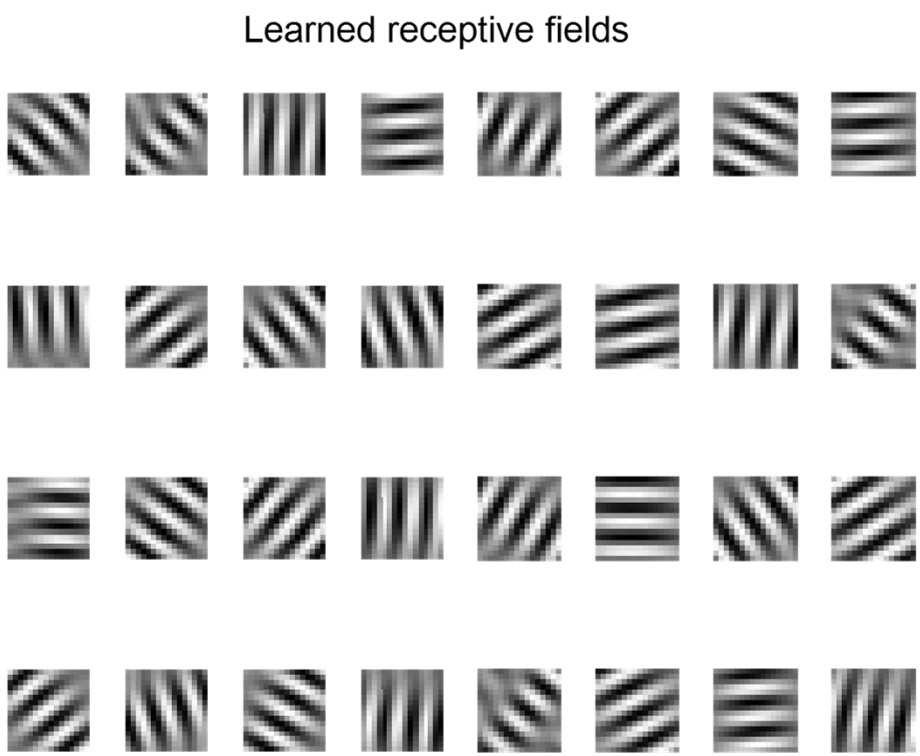
ALWAYS HAVE A FAST WAY TO CHECK YOUR WORK

Initial Verification

Plot the inputs into a single neuron

- Use reshape(W(a,:),16,16) or equivalent

Why do the weight matrices look like the stimuli that they code for?



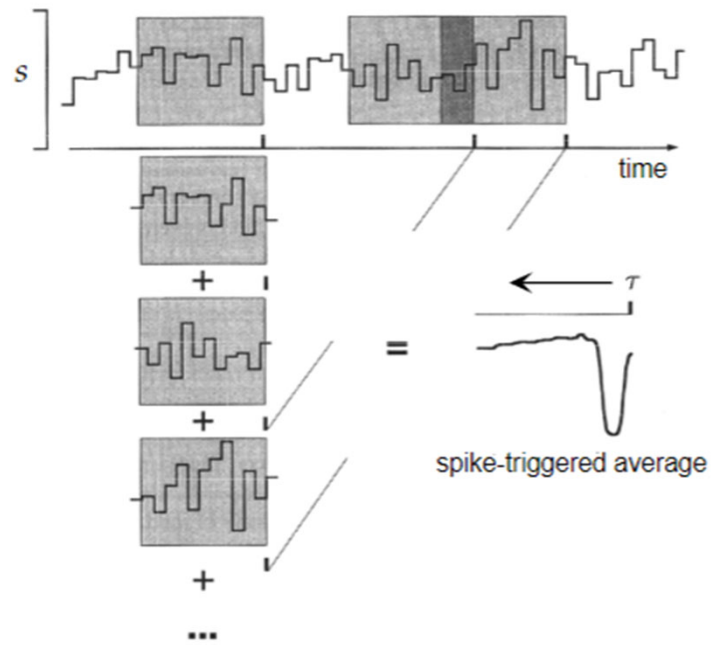
Recall: Determining the receptive field by creating a spike triggered average function

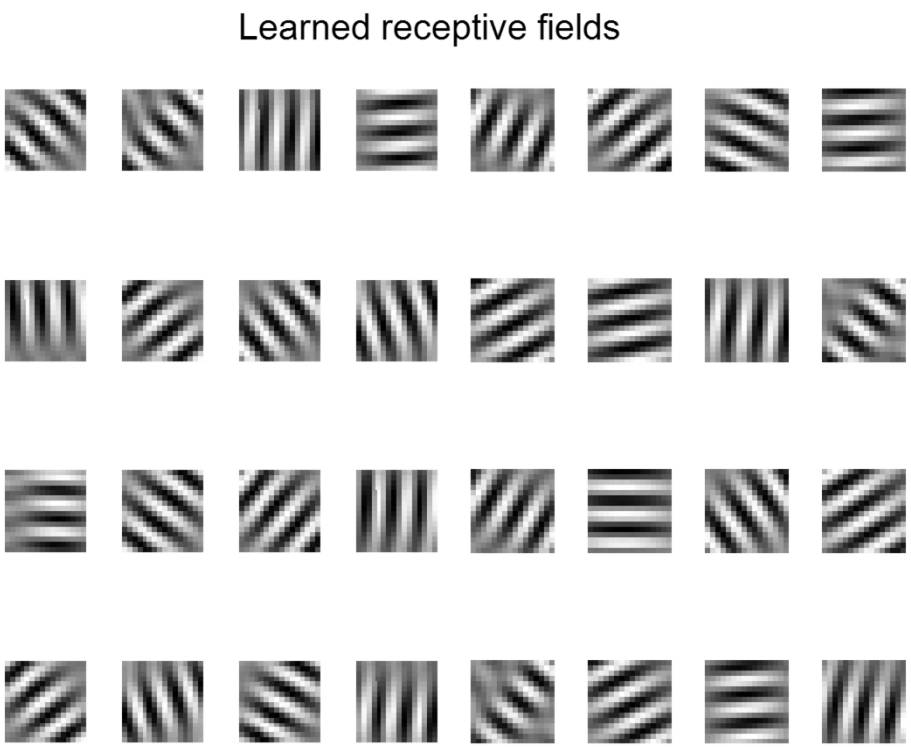
Randomized stimuli create a piecewise function that can be averaged across

$$C(\tau) = < \frac{1}{n} \sum_{i=1}^{n} s(t_i - \tau) >$$

τ acts as a sliding parameter to generate a representation of the 'optimal' stimuli

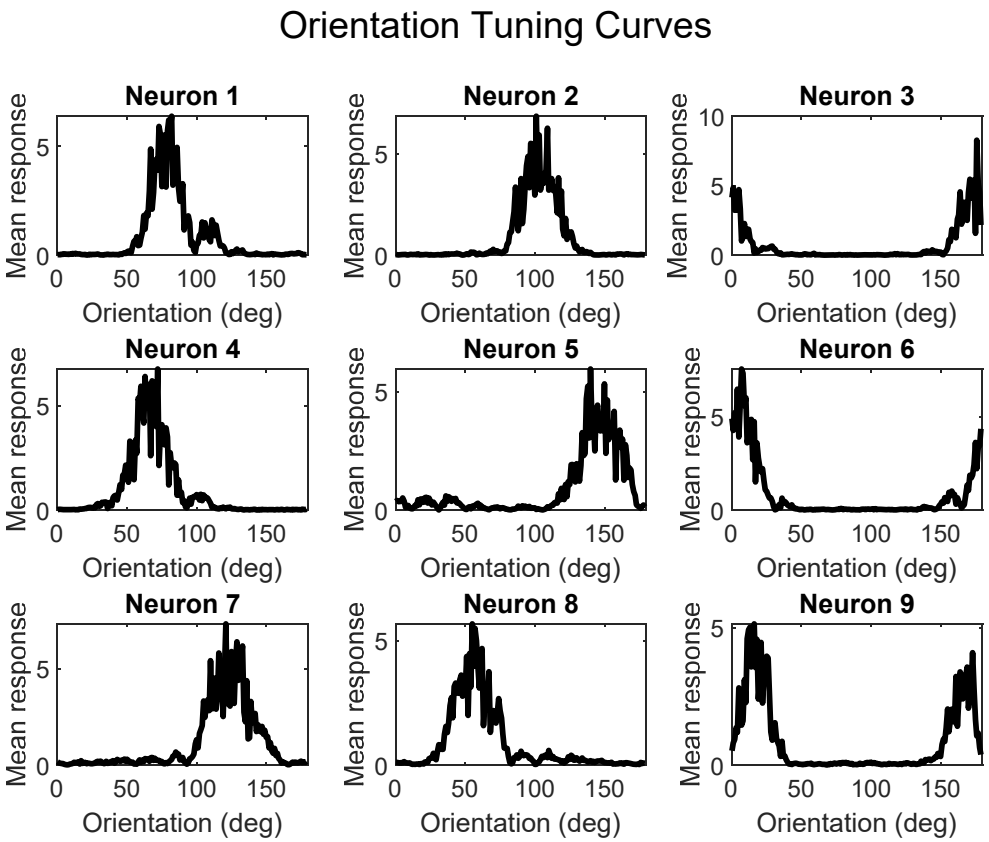
Receptive fields can be thought of as the stimuli that makes the neuron fire maximally





How could we determine the output tuning curves?

WITHOUT ADDITIONALLY SPATIALLY STRUCTURED INITIAL CONDITIONS OR LEARNING RULES, STRUCTURE TENDS TO BE RANDOM



Brute Force Mapping

Step through from 0 to 179 degrees by 1

Fit a gaussian to these to determine mean and standard deviation

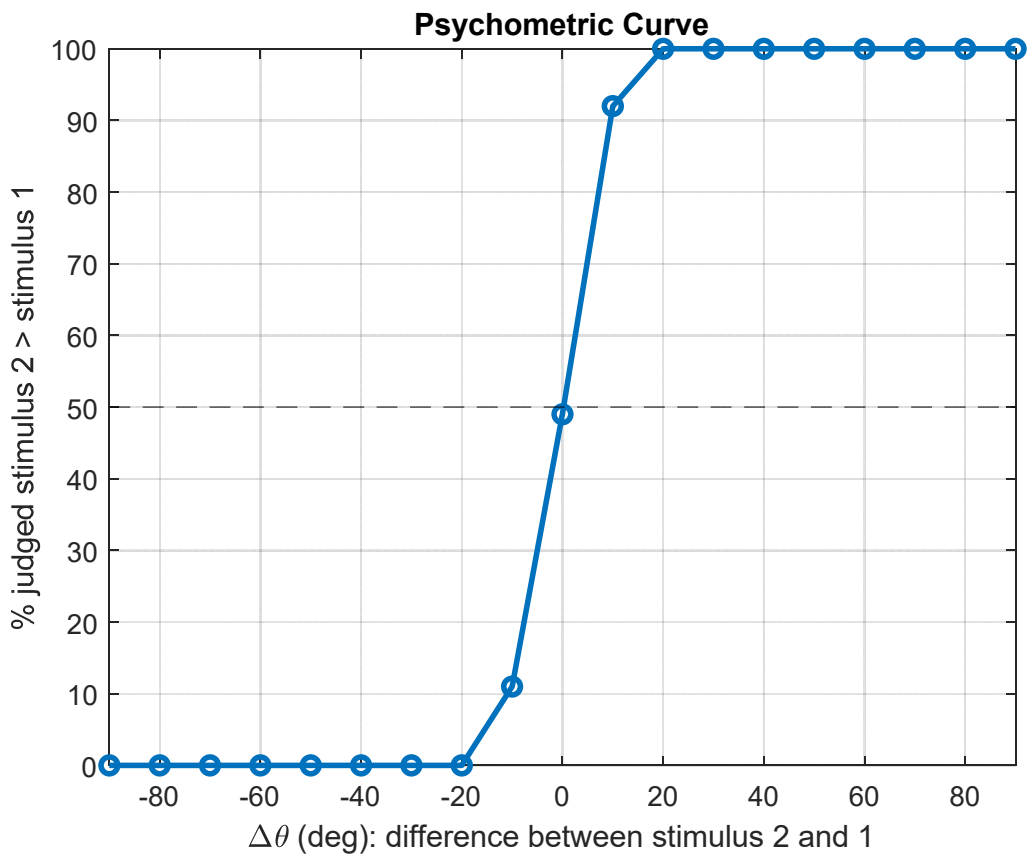
Population Decoding

Using the means of the gaussian, you can have each neuron 'vote' on a preferred orientation (see Problem Set 1)

$$\vec{V} = \sum_a v_a \cdot [\cos(2\mu_a), \sin(2\mu_a)]$$

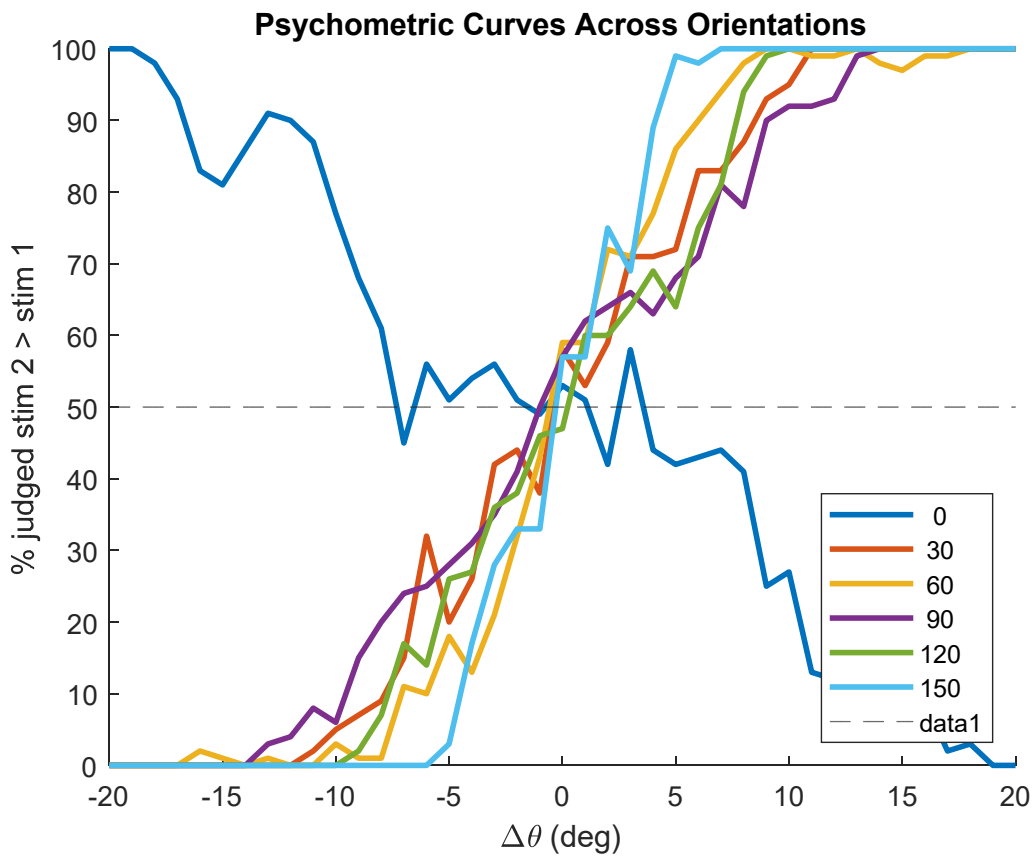
Be sure to keep in mind issues with restraining from [0 180)

$$\theta_{\text{decoded}} = \frac{1}{2} \cdot \tan^{-1}\left(\frac{\sum_a v_a \cdot \sin(2\theta_a)}{\sum_a v_a \cdot \cos(2\theta_a)}\right)$$



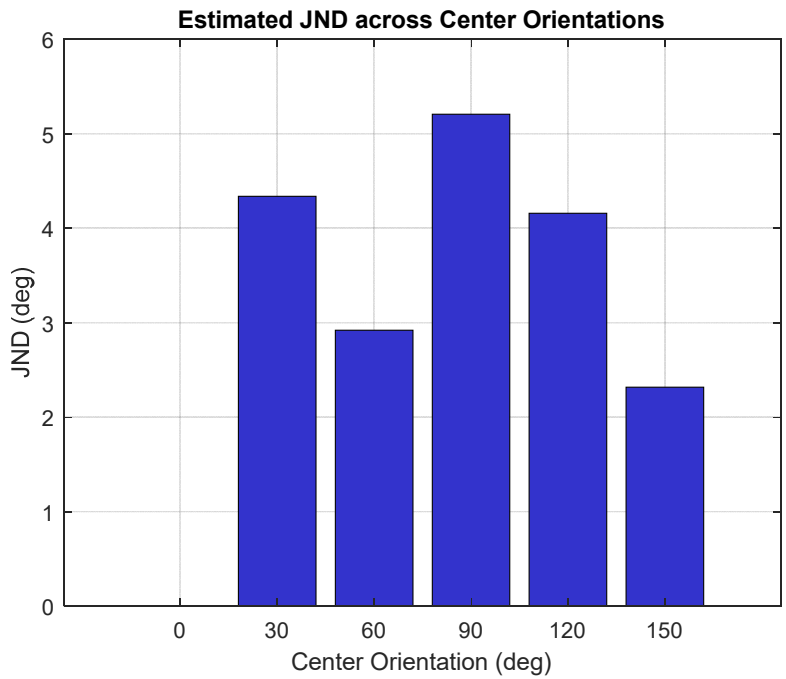
Building a Psychometric Curve

PICK A TEST CASE



Comparing across different angles

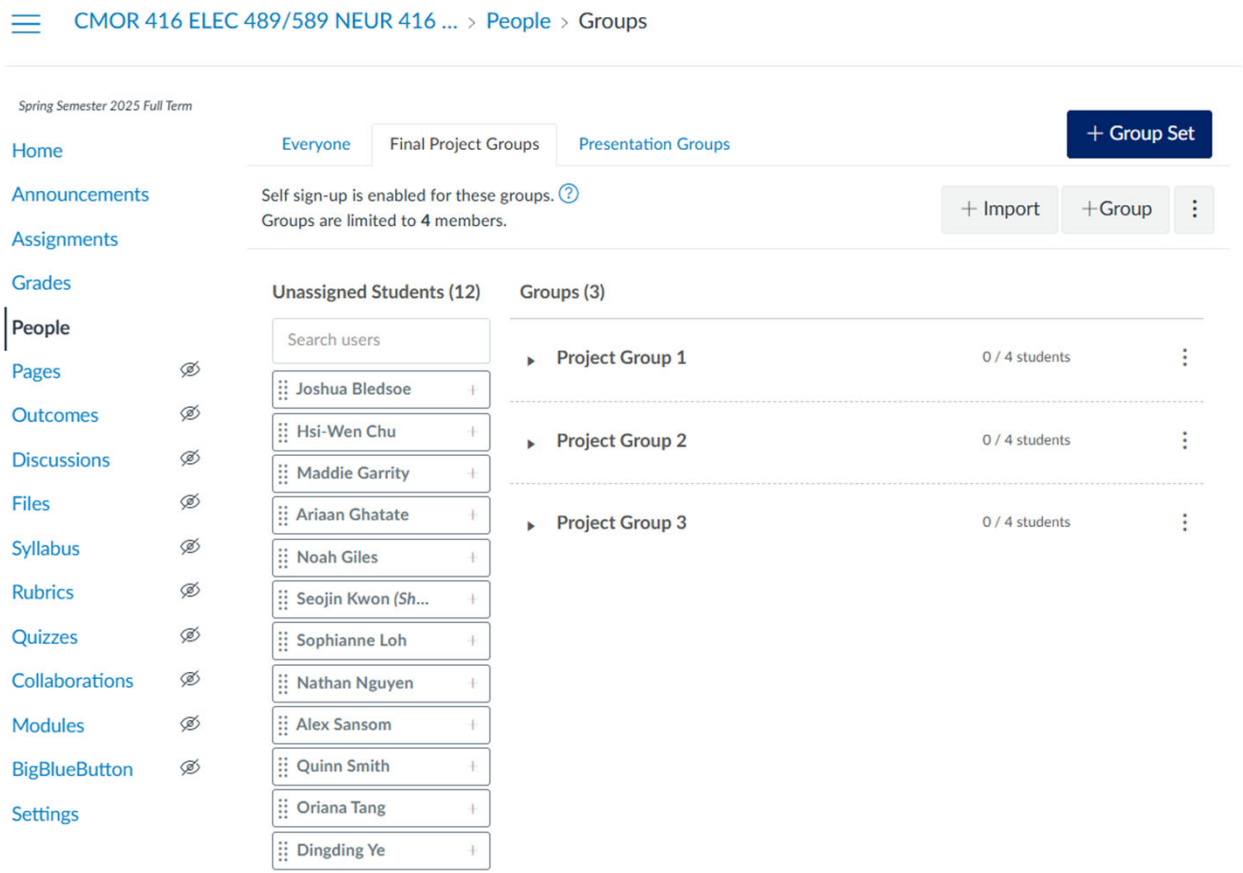
Comparing the JNDs



How this is going to work

PICKING GROUPS TODAY AND TOMORROW

SETTLING ROLES BY 4/30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Spring Semester 2025 Full Term
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Assignments
Grades
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Collaborations
Modules
BigBlueButton
Settings
Everyone
Final Project Groups
Presentation Groups
+ Group Set
Self sign-up is enabled for these groups.
Groups are limited to 4 members.
+ Import
+ Group
Unassigned Students (12)
Groups (3)
Search users
Joshua Bledsoe
Hsi-Wen Chu
Maddie Garrity
Ariaan Ghatate
Noah Giles
Seojin Kwon (Sh...
Sophianne Loh
Nathan Nguyen
Alex Sansom
Quinn Smith
Oriana Tang
Dingding Ye
Project Group 1
0 / 4 students
Project Group 2
0 / 4 students
Project Group 3
0 / 4 students

Appointments

WWW.FLYNN.RICE.EDU/APPOINTMENTS

Homework corrections can be turned in for full credit by the last day of exams

A DAY OR TWO LONGER, IF YOU ASK NICELY AHEAD OF TIME

Closing Out

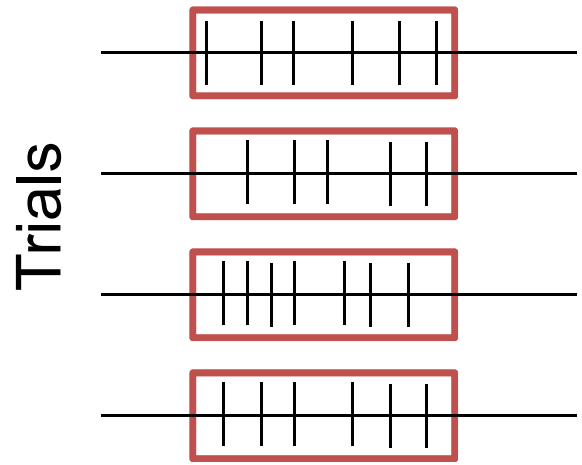
Course Summary

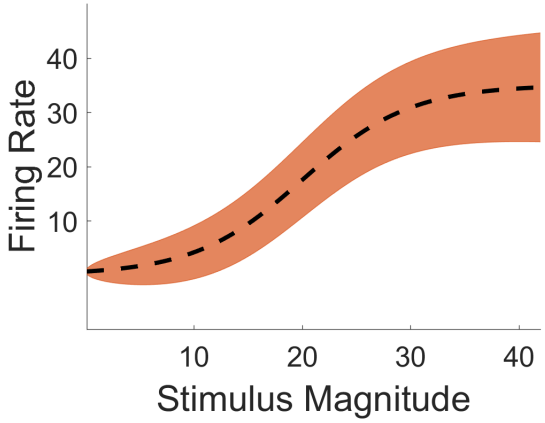
Encoding and Decoding spike trains

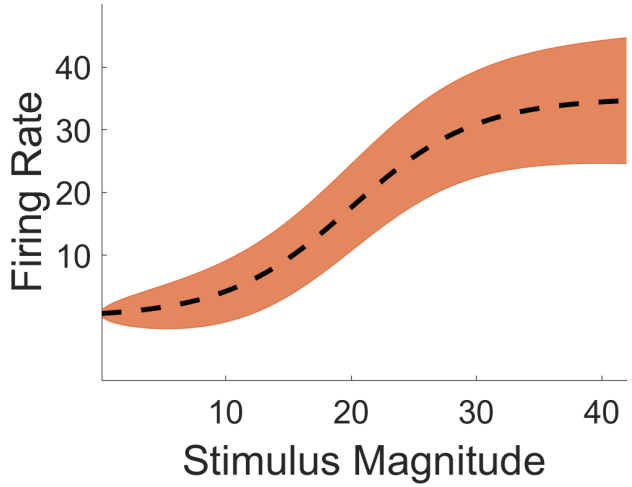
- What does a neural response tell us about a stimulus?

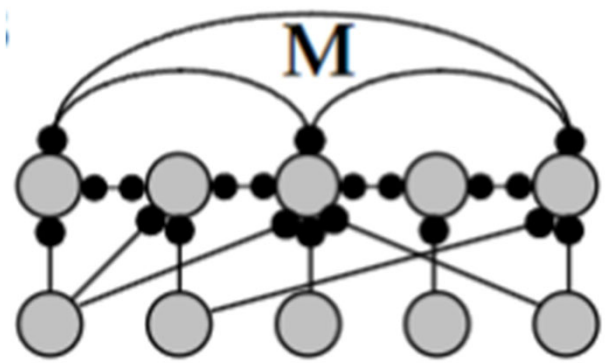
Entropy and Information theory

- How much does the neural response tell us about a stimulus?









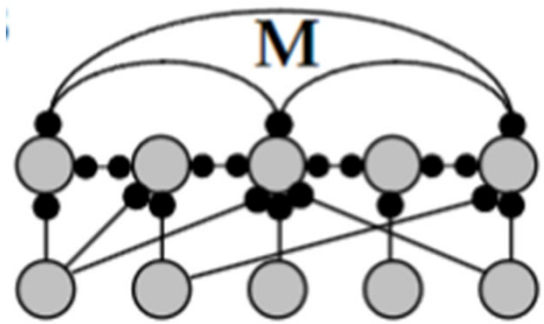
Network models

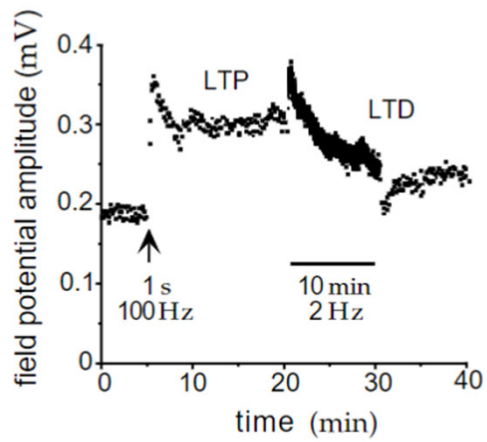
Can we know things about networks without knowing all their details?

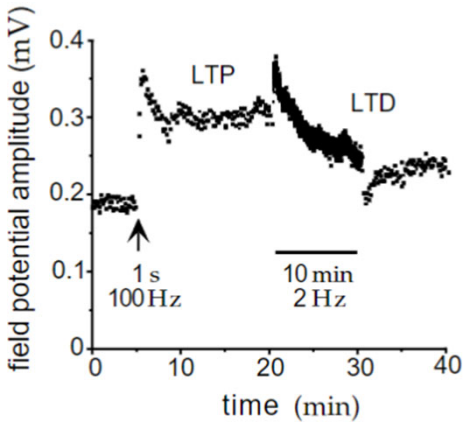
How might networks of neurons

- Coordinate physical movement?
- Maintain short term memory?
- Select attentional focus?

Synaptic Plasticity

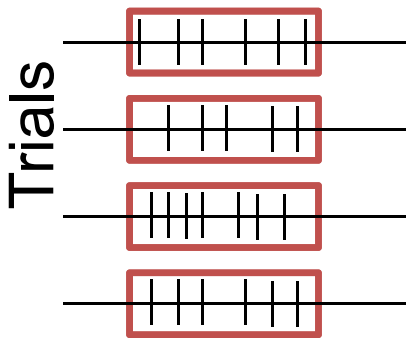


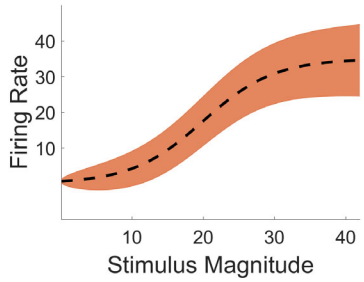


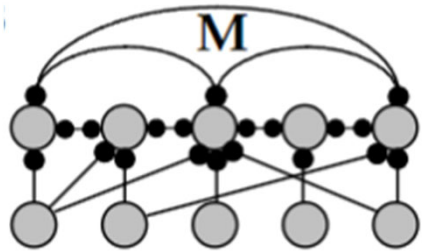


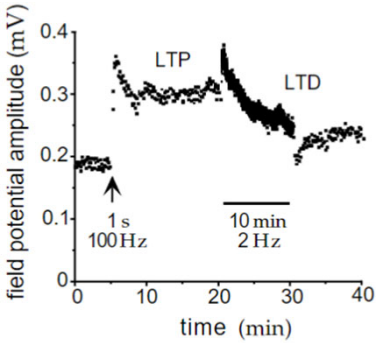
Modeling learning behavior

Jumping layers of abstraction









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